Twist torsion gives executable relation-preserving isogenies

All **eight** cyclic7-kernels from the p827 twist's full7-torsion give certified rational degree7 maps from the source. Each map carries the generator, target and complete signed relation rows with the same scalar. This establishes a concrete connection through standard isogeny theory. It supplies no additional information for the transported base and establishes no solving speedup.

The fixed public source is `icv1-fp10-t5-192cb216`, $y^2=x^3+x+15$ over F827, order823, G=(2,5), Q=37G. Its d2 twist $y^2=x^3+4x+120$ has order833 and structure C7×C119. The two groups have no common nontrivial rational odd-order component. The new construction uses twist torsion as isogeny kernels; it does not add twist-rational points to mixed source relations.

The anti-Frobenius embedding is $(X,Y)\mapsto(X/2,Y\cdot\omega/4)$, $\omega^2=2$. Native checks cover all49 torsion points and all2,401 pairs. Each seven-point line is Frobenius stable. The three paired x-roots construct a normalized Vélu map $\psi(x,y)=(N/H^2,yM/H^3)$, $M=N'H-2NH'$. Exact polynomial curve identities, square-freeness, coprimality, degrees, poles and infinity are checked. Reverse maps come from the image of the full7-torsion, followed by inverse7 coordinate scaling. Both generic compositions equal[7]; inversion on the source subgroup is [588]∘dual since$7\cdot588=1+5\cdot823$.

Line	Quotient $y^2=x^3+a\,x+b$	Canonical model
0	a54,b121	`icv1-fp10-t5-44c1ec1d`
1	a211,b695	`icv1-fp10-t5-bdd22a81`
2	a27,b514	`icv1-fp10-t5-aa4928b9`
3	a11,b543	`icv1-fp10-t5-3b71719e`
4	a229,b473	`icv1-fp10-t5-b4defb92`
5	a17,b601	`icv1-fp10-t5-abb926d8`
6	a665,b221	`icv1-fp10-t5-6c3bd6f1`
7	a367,b162	`icv1-fp10-t5-0f8036c6`

Every quotient has order823. Native generic checks verify16 dual compositions; exhaustive rational checks cover6,584 forward and6,584 reverse points, including identity. Independent PARI constructs all maps and reverse maps with `ellisogeny`, checks exact formulas/compositions and verifies public-point recovery. Eight corrupt-map controls are rejected.

Two fixed target-blind bases contain eight oriented points including G. One uses ascending source coordinates; the other uses shifted coordinates from the lexicographically first maximal-order twist seed, order119. Each mapped base uses the actual image points with preserved signs. All arity2/3 witnesses for Q+jG, j0–15, are enumerated:64 source plus512 destination cases. Every index/sign witness set and resulting coefficient matrix agrees exactly with the corresponding source base. There are1,593 F3 checks,144 explicit identity first-pair exceptions and954 rejected flipped-sign controls.

The ordinary base has40 witness rows, rank9 and verified scalar37. The twist base has66 rows but rank8; the frozen full-system solver returns no complete solution. Counts include repeated-index/sign multiplicities, not distinct independent rows. The separate target-identifiability diagnostic confirms that the twist-base target itself remains ambiguous. All nine models retain each outcome. These are correlated queries for one public target, not18 independent DLPs. No factor-base logs or known37 enter elimination.

Source93a8281dbf29a51ba4be3ba57dce3d6222402bc8; frozen protocol 44d59074bd25ea25 8211b1ef482cfbdd30c96bf8e74e66a4df9a541ef482d153. Native producer and complete native replay pass; independent GP passes strict exit/output/hash checks (only its exact declared stack-size warning). Two whole-panel arms complete. Full protocol/export/file binding has zero findings. Native/CAS bounds are300s each,4GiB,CPU0,one thread. Managed Xeon Platinum8573C KVM, L0; operational timestamps are not performance observations. The documented EKS access/image fallback applies. No paid compute or IAM change.

Four Lean declarations check the structural group implication: `TwistKernelRelations.recovery`, `.injective`, `.relation_iff`, `.scalar_transport`. They assume additive group laws, homomorphisms, the degree7 dual identity and823-torsion. Their proofs establish recovery, injectivity and relation/scalar

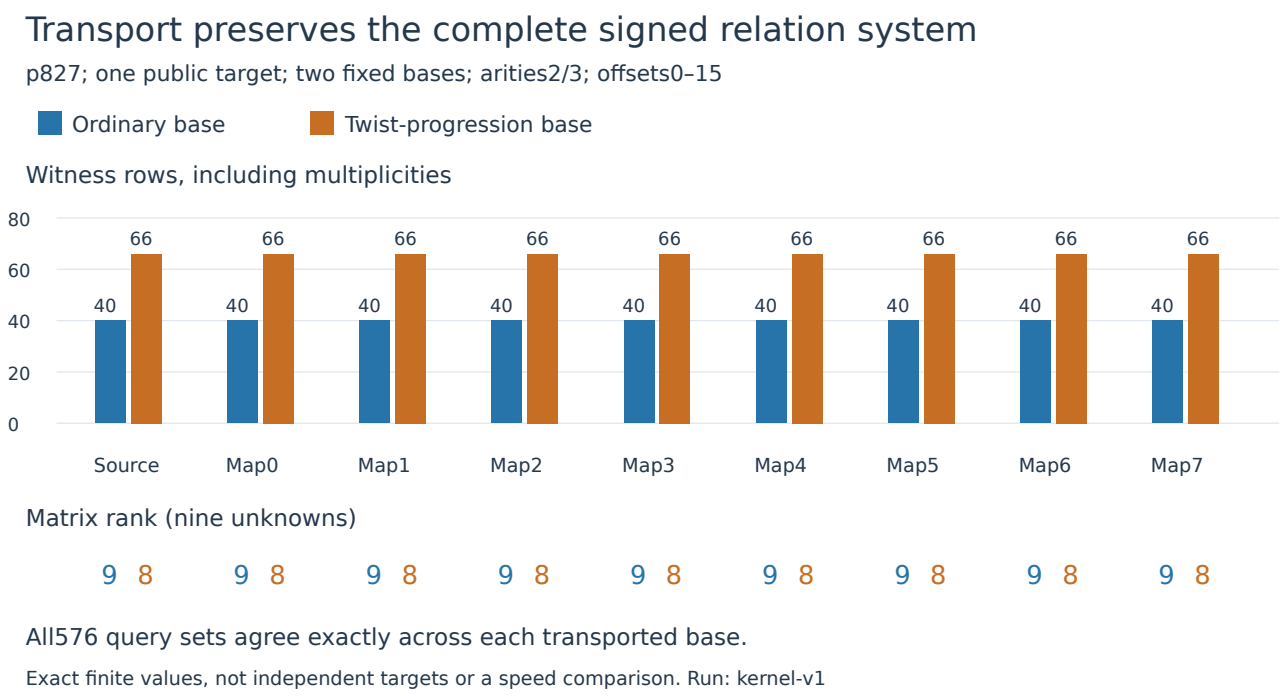


Figure 1: Exact signed rows and matrix ranks

equivalence; the concrete curve/map/dual bridge is **not formalized**. Fresh axiom audit uses only propext, Classical.choice, Quot.sound; no custom axioms, sorry or trusted native evaluation.

The standard descent/Vélu construction is prior art: Sutherland, MIT18.783 Fall2025 Lecture5, Theorems5.11/5.15 and Remarks5.16–17. Primary source and PARI help were read before implementation; atlas isogeny/torsion obstructions were reviewed. Review.

Exact model manifests retain full curve UIDs; ordered routes bind map artifacts and direction. TWK1 identifiers are study map IDs, not a cross-repository ISO1 certification. Eight canonical model records and aliases are registered. No scalar map to the twist's rational group, NIST-size route, novelty, additional transported information, cheapest representative or full IC gain is established. No leaderboard/benchmark row is promoted. Original campaigns and final E28 labels32–63 remain open.

Raw archive:22,024 bytes,10 members, SHA256 58618a66dfec1cbdf743d56e8d0eedde83279bb3b9906adea1710b89f50dd768, in the user-confirmed Setec content-addressed store. See reproduction, admission, export and archive verification.

Next: separately freeze a cold complete decomposition cost panel on these certified maps, with discovery/conversion charged, source/destination same-base solver controls and ordinary group controls. Representation choice may change equation-solving cost; exact transport alone leaves the relation information unchanged. Existing canonical performance graphs stay unchanged because no new performance result was measured; this report adds a correctness diagram.